Graphon Controlled Opinion Dynamics

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Abstract

The code for this paper can be found at the GitHub repository GraphonControl.

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1 The Graph-Limit Dynamics

Let $I := [0, 1]$ be a set of labels. We consider the *graph-limit* (or continuum) opinion dynamics (see [2] for a recent survey), where each agent is indexed by a label $\xi \in I$, and the state of the agent at time t is denoted by $x(t, \xi) \in \mathbb{R}$.

The deterministic dynamics are given by

$$\partial_t x(t,\xi) = \int_I w(t,\xi,\zeta)\,\phi(x(t,\zeta)-x(t,\xi))\,d\zeta, \qquad x \in \mathcal{C}^1\Big([0,T]\to L^2(I;\mathbb{R})\Big) \tag{1}$$

eq:graph_limit

where:

- $w(t,\xi,\zeta)$ is a (possibly non-autonomous) graphon encoding interaction strengths,
- $\phi:\mathbb{R}\to\mathbb{R}$ is the interaction function.

TODO: State two or three examples of real life ϕ with references. Hegelmann-Krauss, bounded confidence, etc.

2 Graphon Controlled Dynamics

We now introduce *control* by allowing the graphon to be time-dependent and chosen by a controller.

TODO: Write the three continuum control problems and the spaces of controls $\mathcal{W}_1,\mathcal{W}_2,\mathcal{W}_3$ explicitly, and discuss the models.

$$\frac{d}{dt}x_i^N = \frac{1}{N}\sum_{j=1}^N \bar{w}_{ij}^N(t,x_i^N(t))\,\phi(x_j^N - x_i^N) + \sigma\,dB_i^t, \quad x_i^N\in\mathbb{R}^d, \quad i\in\{1,\cdots,N\}, \quad t>0$$

where $\{B_i^t\}_{i=1}^N$ are independent and identically distributed Brownian motions. And

$$\bar{w}_{ij}^N(t,x) = N^2\iint_{I_i^N\times I_j^N} w(\xi,\zeta)\,d\xi\,d\zeta, \qquad I_j^N = \left[\frac{j-1}{N},\frac{j}{N}\right).$$

We should think of these three different controls (that mean different things in applications):

$$w(t,\xi,\zeta):\ \text{this is time and label only} \tag{2}$$

$$\bar{w}_{ij}^N(t) = N^2\iint_{I_i^N\times I_j^N} w(t,\xi,\zeta)\,d\xi\,d\zeta, \qquad I_j^N = \left[\frac{j-1}{N},\frac{j}{N}\right).$$

$$w(t,x(\xi),\xi,\zeta):\ \text{this is time, receiver state, and label only} \tag{3}$$

$$\bar{w}_{ij}^N(t,x(\xi)) = N^2\iint_{I_i^N\times I_j^N} w(t,x(\xi),\xi,\zeta)\,d\xi\,d\zeta, \qquad I_j^N = \left[\frac{j-1}{N},\frac{j}{N}\right).$$

$$w(t,x(\xi),x(\zeta),\xi,\zeta):\ \text{this is time, receiver state, sender state, and labels dependent} \tag{4}$$

$$\bar{w}_{ij}^N(t,x(\xi),x(\zeta)) = N^2\iint_{I_i^N\times I_j^N} w(t,x(\xi),x(\zeta),\xi,\zeta)\,d\xi\,d\zeta, \qquad I_j^N = \left[\frac{j-1}{N},\frac{j}{N}\right).$$

3 Cost Functional and Value Function

TODO: Write a paragraph to cue the reader that we are defining the cost functionals to guide the system towards a particular state. Add references to econ/sociology references (Joe: check references Ilan sent).

Our main control objective is to steer the system toward *consensus*. A standard measure of lack of consensus is the variance

$$\mathrm{Var}(x) := \int_I \|x(\xi)-\bar{x}\|^2\,d\xi, \qquad \bar{x} := \int_I x(\xi)\,d\xi.$$

We introduce:

- a running cost $L : \mathcal{X} \times \mathcal{W} \to \mathbb{R}$,
- a terminal cost $G : \mathcal{X} \to \mathbb{R}$.

We not describe an example of a cost which reduces variance and aims at a particular state $x_{target} \in \mathbb{R}$. First, define

$$L(x, w) = \mathrm{Var}(x) + \beta \|w\|^2_{L^2(I^2)}, \qquad G(x) = \lambda \, \mathrm{Var}(x),$$

where $\beta, \lambda \geq 0$. We can also introduce a target state $x_{target} \in \mathbb{R}$, to which we can add to the terminal cost as

$$G(x; x_{target}) = \gamma |\overline{x}(T) - x_{target}|^2. \tag{5}$$

For an admissible control $w(\cdot)$, define the cost

$$J(t, x; w) = \int_t^T L(x(s), w(s)) \, ds + G(x(T)).$$

The associated value function is

$$V(t, x) := \inf_{w(\cdot)} J(t, x; w), \qquad (t, x) \in [0, T] \times \mathcal{X}.$$

We discuss the associated infinite dimensional Hamilton-Jacobi-Bellman equation in the Appendix.

4 Finite Agent System

In this section, we define the N-agent dynamics. Let $\{\xi_i\}_{i=1}^N \subset I$ be a discretization of labels given by $\xi_i = (i - 0.5)/N$. Then, we approximate the controlled dynamics for each $w \in \mathcal{W}_k$ $(k = 1, 2, 3)$ using the histogram function approximation

$$x_N(t, \xi) = \sum_{i=1}^N x_i^N(t) \, \mathbf{1}_{I_i^N}(\xi), \qquad I_i^N = \left[\frac{i-1}{N}, \frac{i}{N}\right), \tag{6}$$

eqn: finite agent approximation

where $\{x_i^N(t)\}_{i=1}^N \subset \mathbb{R}$ is the list of the opinions of N agents. We shall be working in discrete time, with time step $\Delta t > 0$ with the total number of time steps $N_T = T/\Delta t$. We shall discretize the graphon by the piecewise constant graphon

$$w^N(t, \xi, \zeta) = \sum_{i,j=1}^N \overline{w}_{i,j}^N(t) \, \mathbf{1}_{I_i^N}(\xi) \mathbf{1}_{I_j^N}(\zeta), \tag{7}$$

eqn: finite graphon approximation A

with the mean piecewise constant interaction by

$$\overline{w}_{i,j}^N(t) = N^2 \int_{I_i^N \times I_j^N} w(t, \xi, \zeta) d\zeta \, d\xi. \tag{8}$$

eqn: finite graphon approximation B

Now note we can consider the above equation in terms of a function of the discretized labels $\xi_i, \zeta_j \in I$ given by

$$\overline{w}^N(t, \xi_i, \zeta_j) = \overline{w}_{i,j}^N(t). \tag{9}$$

eqn: finite graphon approximation C

4.1 Time and Label Dependent Controls

In this subsection, we will discretize the dynamics for time and label only controls $w \in \mathcal{W}_1$.

4.1.1 Dynamics

We will now describe the N-agent system by approximating solutions of the continuum equation

$$\partial_t x(t,\xi) = \int_I w(t,\xi,\zeta)\,\phi\big(x(t,\zeta) - x(t,\xi)\big)\,d\zeta \tag{10}$$

by using eq. (6). This results in the dynamics for each time step $n = 0, \ldots, N_T - 1$, with $s_n = n\Delta t$ and

$$x_i^N(s_n) = x_i^N(s_{n-1}) + \frac{\Delta t}{N}\sum_{j=1}^{N} \overline{w}^N(s_{n-1},\xi_i,\zeta_j)\phi(x_j^N(s_{n-1}) - x_i^N(s_{n-1})) + \sigma\Delta B_{n-1}^i. \tag{11}$$

Here $\sigma > 0$ is the opinion variance and $\Delta B_{n-1}^i \sim \mathcal{N}(0,\Delta t)$ are i.i.d samples from the Gaussian distribution with mean 0 and variance Δt.

4.1.2 Discretization of the Cost

Let $w = w_\theta$ be an approximation of the control via deep neural network which we will define in Section 5.1. Then define the parametrized cost

$$J(\theta) = \int_0^T L(x(s), w_\theta(s,\cdot,\cdot))\,ds + G(x(T); x_{target})$$

and the N-agent parametrized cost

$$\begin{aligned} J^N(\theta) &= \int_0^T \mathrm{Var}(x_N(s)) + \beta\,\|w_\theta(s)\|^2_{L^2(I^2)}\,ds + \lambda\mathrm{Var}(x_N(T)) + \gamma|\overline{x}_N(T) - x_{target}|^2 \\ &= \int_0^T \left[\int_I \big|x_N(s,\xi) - \bar{x}_N(s))\big|^2\,d\xi + \beta R_\theta(s)\right] ds \\ &\quad + \lambda\int_I |x_N(T,\xi) - \overline{x}_N(T,\xi)|^2 d\xi + \gamma|\overline{x}_N(T) - x_{target}|^2 \end{aligned}$$

Let's explicitly compute this in terms of the trajectories of the individual agents. Fix $s \in [0,T]$, and first compute average as

$$\bar{x}_N(s) = \int_I x_N(s,\xi)\,d\xi = \sum_{i=1}^N \int_{I_i^N} x_i^N(s)\,d\xi = \frac{1}{N}\sum_{i=1}^N x_i^N(s).$$

For the variation we have

$$\begin{aligned} \int_I \big|x_N(s,\xi) - \bar{x}_N(s)\big|^2\,d\xi &= \int_I x_N^2(s,\xi) + \bar{x}_N^2(s) - 2x_N(s,\xi)\bar{x}_N(s)\,d\xi \\ &= \int_I x_N^2(s,\xi) - 2\bar{x}_N(s)\int_I x_N(s,\xi)\,d\xi + \bar{x}_N^2(s) \\ &= \int_I x_N^2(s,\xi) - \bar{x}_N^2(s) \end{aligned}$$

By using eq. (6), we can write this in terms of the ensemble of agents as

$$\begin{aligned}
\int_I \left|x_N(s,\xi) - \bar{x}_N(s)\right|^2 d\xi &= \int_I \sum_{i=1}^N \sum_{j=1}^N \mathbf{1}_{I_i^N}(\xi)\mathbf{1}_{I_j^N}(\xi) x_i^N(s) x_j^N(s)\, d\xi - \bar{x}_N^2(s) \\
&= \int_I \sum_{k=1}^N \mathbf{1}_{I_k^N}(\xi)(x_k^N(s))^2\, d\xi - \bar{x}_N^2(s) \\
&= \sum_{k=1}^N \int_{I_k^N} (x_k^N(s))^2\, d\xi - \bar{x}_N^2(s) \\
&= \frac{1}{N}\sum_{k=1}^N (x_k^N(s))^2 - \bar{x}_N^2(s)
\end{aligned}$$

Since

$$\bar{x}_N^2(s) = \left(\frac{1}{N}\sum_{i=1}^N x_i^N(s)\right)^2 = \frac{1}{N^2}\left(\sum_{i=1}^N x_i^N(s)\right)^2 = \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(s)x_j^N(s)$$

we can explicitly compute the variance of the N-agent system as

$$\mathrm{Var}(x_N(s)) = \frac{1}{N}\sum_{k=1}^N (x_k^N(s))^2 - \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(s)x_j^N(s)$$

Putting these computations together, we obtain

$$\begin{aligned}
J^N(\theta) = &\int_0^T \left[\frac{1}{N}\sum_{k=1}^N (x_k^N(s))^2 - \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(s)x_j^N(s)\right] ds \\
&+ \beta \int_0^T \|w_\theta(s)\|^2_{L^2(I^2)}\, ds \\
&+ \lambda\left[\frac{1}{N}\sum_{k=1}^N (x_k^N(T))^2 - \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(T)x_j^N(T)\right] \\
&+ \gamma|\overline{x}_N(T) - x_{target}|^2
\end{aligned}$$

And we can also get the approximation to discretize time,

$$\begin{aligned}
J^{N,\Delta t}(\theta) = &\sum_{n=0}^{N_T-1} \left[\frac{1}{N}\sum_{k=1}^N (x_k^N(s_n))^2 - \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(s_n)x_j^N(s_n)\right]\Delta t \\
&+ \frac{\beta}{2}\sum_{n=0}^{N_T-1} \|w_\theta(s_n)\|^2_{L^2(I^2)}\Delta t \qquad \text{eqn:J-cost-1} \\
&+ \lambda\left[\frac{1}{N}\sum_{k=1}^N (x_k^N(T))^2 - \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N x_i^N(T)x_j^N(T)\right] \\
&+ \gamma|\overline{x}_N(T) - x_{target}|^2
\end{aligned} \tag{12}$$

4.2 Time, Label, and Local State Dependent Controls

In this subsection, we will discretize the dynamics for time, label and local state dependent controls $w \in \mathcal{W}_2$. TODO

4.2.1 Dynamics

For the time, label, and receiver state only control ($w \in \mathcal{W}_2$), we will discretize the dynamics as

$$x_i^N(s_n) = x_i^N(s_{n-1}) + \frac{\Delta t}{N} \sum_{j=1}^{N} w\Big(s_{n-1}, x_i^N(s_{n-1}), \xi_i, \xi_j\Big) \phi(x_j^N(s_{n-1}) - x_i^N(s_{n-1})) + \sigma \Delta B_{n-1}^i. \tag{13}$$

where $\sigma > 0$ is the opinion variance and $\Delta B_{n-1}^i \sim \mathcal{N}(0, \Delta t)$ are i.i.d samples from the Gaussian distribution with mean 0 and variance Δt.

4.2.2 Discretization of the Cost

TODO

4.3 Time, Label, and Sender/Receiver State Dependent Controls

In this subsection, we will discretize the dynamics for time, label, and Sender/Reciever state dependent controls $w \in \mathcal{W}_3$. TODO

4.3.1 Dynamics

For the full time, label, receiver state, and sender state control ($w \in \mathcal{W}_3$), we have

$$x_i^N(s_n) = x_i^N(s_{n-1}) + \frac{\Delta t}{N} \sum_{j=1}^{N} w\Big(s_{n-1}, x_i^N(s_{n-1}), x_j^N(s_{n-1}), \xi_i, \xi_j\Big) \phi(x_j^N(s_{n-1}) - x_i^N(s_{n-1})) + \sigma \Delta B_{n-1}^i, \tag{14}$$

where $\sigma > 0$ is the opinion variance and $\Delta B_{n-1}^i \sim \mathcal{N}(0, \Delta t)$ are i.i.d samples from the Gaussian distribution with mean 0 and variance Δt. TODO

4.3.2 Discretization of the Cost

TODO

5 Neural Network Architectures

sec:nn-arch

5.1 Time and Label Dependent Controls

subsec:nn-arch-1

We now describe a neural parameterization of the interaction kernel in which the control depends on time and labels only (and *not* on the current opinion state). In particular, we parameterize a time-dependent graphon

$$w_\theta : [0, T] \times I^2 \to [0, 1], \qquad (t, \xi, \zeta) \mapsto w_\theta(t, \xi, \zeta). \tag{15}$$

eqn:nn-def-labelonly

The controlled dynamics remain state-dependent through the evolution operator (cf. (31)), but the controller itself is *label-only*.

Constraints. We impose the structural constraints

$$0 \le w_\theta(t, \xi, \zeta) \le 1, \tag{16}$$

eqn:nn-range-labelonly

$$w_\theta(t, \xi, \xi) = 0 \quad \text{for all } \xi \in I, \tag{17}$$

eqn:nn-noself-labelonly

$$\int_I w_\theta(t, \xi, \zeta)\, d\zeta = 1 \quad \text{for all } (t, \xi) \in [0, T] \times I, \tag{18}$$

eqn:nn-rownorm-labelonly

which encode a fixed attention budget per receiver label.

Discretization. We discretize $I = [0,1]$ using nodes $\{\xi_i\}_{i=1}^N \subset I$ and represent the kernel by an $N \times N$ matrix $W_\theta(t)$ with entries

$$W_{\theta,ij}(t) \approx \frac{1}{N}\overline{w}_{i,j}(t) = \frac{1}{N}\overline{w}^N(t,\xi_i,\xi_j) \tag{19}$$

eqn:nn-discrete-labelonly

Feature maps and scoring. To represent $\xi \in I$ we use a Fourier feature map $\psi : I \to \mathbb{R}^{2m}$ (e.g. $\psi(\xi) = (\sin(2\pi 2^k \xi), \cos(2\pi 2^k \xi))_{k=0}^{m-1}$). Define receiver and sender embeddings

$$h_i := \phi_\theta\big(\psi(\xi_i)\big) \in \mathbb{R}^d, \qquad \widetilde{h}_j := \widetilde{\phi}_\theta\big(\psi(\xi_j)\big) \in \mathbb{R}^d \tag{20}$$

eqn:nn-embed-labelonly

where $\phi_\theta, \widetilde{\phi}_\theta : \mathbb{R}^{2m} \to \mathbb{R}^d$ are a feed-forward neural network. We then define scores

$$s_{ij}(t) := g_\theta\big(t, h_i, \widetilde{h}_j\big) \in \mathbb{R}, \tag{21}$$

eqn:nn-score-labelonly

where $g_\theta : \mathbb{R}^{1+2d} \to \mathbb{R}$ is a feed-forward neural network, and we mask self-interactions by setting $s_{ii}(t) = -\infty$ (in practice, a large negative number).

Row normalization. Finally, we obtain the weight matrix via a masked row-softmax:

$$W_{\theta,ij}(t) := \frac{\exp(s_{ij}(t))\,\mathbf{1}_{\{i\neq j\}}}{\sum_{k\neq i}\exp(s_{ik}(t))}, \qquad 1 \le i,j \le N. \tag{22}$$

eqn:nn-softmax-labelonly

By construction, $W_{\theta,ij}(t) \ge 0$, $W_{\theta,ii}(t) = 0$, and $\sum_{j=1}^N W_{\theta,ij}(t) = 1$.

5.2 Time, Label, and Local State Dependent Controls

subsec:nn-arch-2

We now describe the neural parameterization of the interaction kernel. In the model studied in this paper, the weight assigned by agent i (with label $\xi_i \in I$ and local state $x_i \in \mathbb{R}$) to agent j (with label $\xi_j \in I$) at time $t \in [0,T]$ is given by

$$w_\theta : [0,T] \times \mathbb{R} \times I^2 \to [0,1], \qquad (t,x,\xi,\zeta) \mapsto w_\theta(t,x,\xi,\zeta), \tag{23}$$

eqn:nn-def-local

and we interpret

$$W_{\theta,ij}(t) \approx w_\theta\big(t, x_i(t), \xi_i, \xi_j\big).$$

In particular, the kernel depends on the *local* (receiver) state $x_i(t)$ and on the pair of labels (ξ_i, ξ_j), but not on the sender state $x_j(t)$.

Constraints. We recall the following structural constraints:

$$0 \le w_\theta(t,x,\xi,\zeta) \le 1, \tag{24}$$

eqn:nn-range-local

$$w_\theta(t,x,\xi,\xi) = 0 \quad \text{for all } (t,x,\xi) \in [0,T] \times \mathbb{R} \times I, \tag{25}$$

eqn:nn-noself-local

$$\int_I w_\theta(t,x,\xi,\zeta)\,d\zeta = 1 \quad \text{for all } (t,x,\xi) \in [0,T] \times \mathbb{R} \times I. \tag{26}$$

eqn:nn-rownorm-local

Condition (26) corresponds to a fixed "attention budget" for each receiver label ξ. We *do not* enforce symmetry $w_\theta(t,x,\xi,\zeta) = w_\theta(t,x,\zeta,\xi)$ nor column normalization; this allows heterogeneous in-degrees (e.g. "influencers") and directed exposure patterns, which is natural for recommendation systems.

Discretization. We discretize $I = [0,1]$ using nodes $\{\xi_i\}_{i=1}^N \subset I$ and represent the kernel by an $N \times N$ matrix $W_\theta(t,x)$ with entries

$$W_{\theta,ij}(t) := w_\theta\big(t, x_i(t), \xi_i, \xi_j\big), \qquad i,j \in \{1,\dots,N\}. \tag{27}$$

eqn:nn-discrete-local

Feature maps. Each receiver node (x_i, ξ_i) is mapped to an embedding

$$h_i = \phi_\theta\big(t, x_i, \xi_i\big) \in \mathbb{R}^d, \tag{28}$$

eqn:nn-embed-local

where ϕ_θ is a feed-forward neural network. To represent $\xi \in I$ we use a Fourier feature map $\psi : I \to \mathbb{R}^{2m}$ (e.g. $\psi(\xi) = (\sin(2\pi 2^k \xi), \cos(2\pi 2^k \xi))_{k=0}^{m-1}$).

Row-wise scoring without sender state. To ensure that $W_{\theta,ij}$ depends on the sender only through its label ξ_j (and not through x_j), we define a row-wise score function

$$s_{ij} := g_\theta\big(h_i, \psi(\xi_j)\big) \in \mathbb{R}, \tag{29}$$

eqn:nn-score-local

where $g_\theta : \mathbb{R}^{d+2m} \to \mathbb{R}$ is a feed-forward neural network. We then mask self-interactions by setting $s_{ii} = -\infty$ (in practice a large negative number).

Row normalization. The weight matrix is obtained by a masked row-softmax:

$$W_{\theta,ij}(t,x) := \frac{\exp(s_{ij})\,\mathbf{1}_{\{i \neq j\}}}{\sum_{k \neq i} \exp(s_{ik})}, \qquad 1 \le i, j \le N. \tag{30}$$

eqn:nn-softmax-local

By construction,

$$W_{\theta,ij}(t,x) \ge 0, \qquad W_{\theta,ii}(t,x) = 0, \qquad \sum_{j=1}^{N} W_{\theta,ij}(t,x) = 1.$$

Thus (24)–(26) hold at the discrete level, and the approximation (27) yields a differentiable parameterization suitable for gradient-based optimization.

rem:row-stoch

Remark 1 (Modeling implications of row-stochasticity)**.** *The normalization* (26) *(equivalently* $\sum_j W_{\theta,ij} = 1$*) encodes a finite attention budget: each agent distributes one unit of attention across the population at each time. Dropping column normalization permits unequal total exposure across agents and allows hubs or influencers to emerge. Dropping symmetry permits directed influence and asymmetric recommendation patterns, both of which are typical in social platforms.*

5.3 Time, Label, and Sender/Reciever State Controls

subsec:nn-arch-3

TODO: Later, keep paragraphs consistent with the above.

6 Numerical Results

6.1 $\mathcal{W}_1$ Control Experiments

6.2 $\mathcal{W}_2$ Control Experiments

6.3 $\mathcal{W}_3$ Control Experiments

7 Appendix

sec: Appendix

In this Appendix, we discuss the Hamilton-Jacobi-Bellman formulation of the problem (1). For $x \in \mathcal{X}$ and $w \in \mathcal{W}$, define the nonlinear operator

$$\big(F(x,w)\big)(\xi) := \int_I w(\xi,\zeta)\,\phi(x(\zeta)-x(\xi))\,d\zeta.$$

The controlled dynamics can then be written as an abstract ODE on $\mathcal{X}$:

$$\dot{x}(t) = F(x(t),w(t)), \qquad x(0) = x_0 \in \mathcal{X}. \tag{31}$$

eq:abstract_dynamics

7.1 The Hamilton–Jacobi–Bellman Equation

Formally, the value function V satisfies an infinite-dimensional Hamilton–Jacobi–Bellman (HJB) equation.

Let $D_xV(t,x)$ denote the Fréchet (or Gâteaux) derivative of V with respect to the state x. The duality pairing with an element $f \in \mathcal{X}$ representing an infinitesimal variation of the state is

$$\langle D_xV(t,x), f\rangle = \int_I D_xV(t,x)(\xi)\cdot f(\xi)\,d\xi.$$

7.2 Derivation

Choose a small $\Delta t > 0$ such that

$$\int_t^T \cdot = \int_t^{t+\Delta t} \cdot + \int_{t+\Delta t}^T \cdot$$

So, the cost function can be written

$$J(t,x;w) = \int_t^{t+\Delta t} L(x(s),w(s))\,ds + \int_{t+\Delta t}^T L(x(s),w(s))\,ds + G(x(T))$$

$$V(t+\Delta t, x(t+\Delta t)) = \inf_{w \text{ for } [t+\Delta t,T]} \left\{ \int_{t+\Delta t}^T L(x(s),w(s))\,ds + G(x(T)) \right\}$$

So,

$$V(t,x) = \inf_{w \text{ for } [t,t+\Delta t]} \left\{ \int_t^{t+\Delta t} L(x(s),w(s))\,ds + V(t+\Delta t, x(t+\Delta t)) \right\}$$

and

$$\int_t^{t+\Delta t} L(x(s),w(s))\,ds \approx L(x(s),w(s))\Delta t$$

where

$$x(t+\Delta t) \approx x(t) + \Delta t F(x(t),w(t))$$

so

$$V(t,x) = \inf_{w\in\mathcal{W}} \left\{ L(x(s),w(s))\Delta t + V(t+\Delta t, x+\Delta t F(x,w)) \right\}$$

Apply Taylor expansion to $V(t+\Delta t, x+\Delta t F)$, we have

$$V(t+\Delta t, x+\Delta t F) = V(t, x+\Delta t F) + \partial_t V(t, x+\Delta t F)\Delta t$$

Apply Taylor expansion to $V(t, x+\Delta t F)$, we have

$$V(t, x+\Delta t F) = V(t,x) + D_xV(t,x)(\Delta t F) = V(t,x) + \Delta t D_xV(t,x)(F)$$

Combine together,

$$\begin{aligned} V(t+\Delta t, x+\Delta F) &= V(t,x) + \Delta t D_x V(t,x)(F) + \Delta t \partial_t V(t, x+\Delta t F) \\ &= V(t,x) + \Delta t \langle D_x V(t,x), F\rangle + \Delta t \partial_t V(t, x+\Delta t F) \end{aligned}$$

then we get

$$V(t,x) = \inf_w \Big\{ L(x,w)\Delta t + V(t,x) + \Delta t \langle D_x V(t,x), F\rangle + \Delta t \partial_t V(t, x+\Delta t F) \Big\}$$

And finally, the HJB equation reads

$$-\partial_t V(t,x) = \inf_{w\in\mathcal{W}} \Big\{ L(x,w) + \langle D_x V(t,x), F(x,w)\rangle \Big\}, \qquad V(T,x) = G(x). \tag{32}$$

eq:HJB

Expanding the pairing using the definition of F,

$$\langle D_x V, F(x,w)\rangle = \int_I\!\!\int_I D_x V(t,x)(\xi) \cdot \big[w(\xi,\zeta)\, \phi(x(\zeta) - x(\xi)) \big]\, d\zeta\, d\xi.$$

7.3 Hamiltonian and Closed-Loop Control

Define the Hamiltonian

$$\mathcal{H}(x,p) := \inf_{w\in\mathcal{W}} \Big\{ L(x,w) + \int_I\!\!\int_I p(\xi) \cdot \big[w(\xi,\zeta)\, \phi(x(\zeta) - x(\xi)) \big]\, d\zeta\, d\xi \Big\},$$

where $p \in \mathcal{X}$.

Then the HJB equation (32) can be written compactly as

$$-\partial_t V(t,x) = \mathcal{H}\big(x, D_x V(t,x)\big).$$

Provided V is sufficiently regular, the optimal *closed-loop* (control-in-feedback-form) graphon is given by

$$w^*(t,\cdot,\cdot) = \arg\min_{w\in\mathcal{W}} \Big\{ L(x,w) + \langle D_x V(t,x), F(x,w)\rangle \Big\}\Big|_{x=x(t)}.$$

This abstract formulation is the starting point for:

- adjoint/Pontryagin methods when ϕ is known,
- gradient-free or learning-based approximations when ϕ is unknown.

References

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